

Dr. Isah Charles Saidu

Lead Instructor

Liatris Systems

AI & Technology Solutions

Session Duration: 1 Hour

July 5, 2026



Liatris Systems



In partnership with the Independent National Electoral Commission

Meet Your Instructor



Dr. Isah Charles Saidu

-  Lead Instructor, INEC AI Programme
-  **Liatris Systems** – Founder & CEO
-  **Baze University** – Senior Faculty
-  PhD in Computer Science
-  AI & Machine Learning Specialist
-  Over 15 years of experience
-  Training ICT professionals across Nigeria

Welcome Message

"I am honoured to facilitate this capacity building programme. My goal is to equip you with practical AI skills that will directly support INEC's mission of delivering credible elections through technology."

Session Roadmap

Part 1: Foundations

-  What is Artificial Intelligence?
-  A Brief History of AI
-  Types of AI

Part 2: Machine Learning

-  What is Machine Learning?
-  Types of Machine Learning
-  The AI Project Lifecycle

Part 3: AI in Elections

-  AI Applications at INEC
-  Use Cases and Benefits

Part 4: Responsible AI

-  Ethics and Governance
-  Key Takeaways
-  Q&A Session

Session Objective: Understand how AI can transform election management

What is Artificial Intelligence?

What is Artificial Intelligence?

Artificial Intelligence (AI)

is the simulation of human intelligence in machines that are programmed to think, learn, and make decisions.

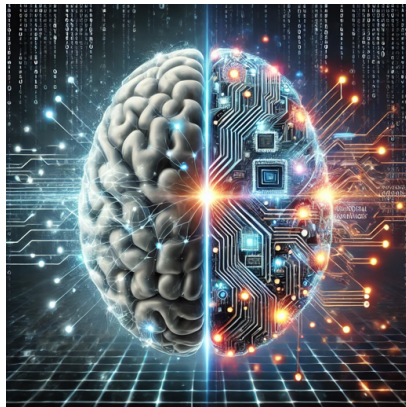
Key Capabilities

👁 Perception

⚙ Reasoning

🎓 Learning

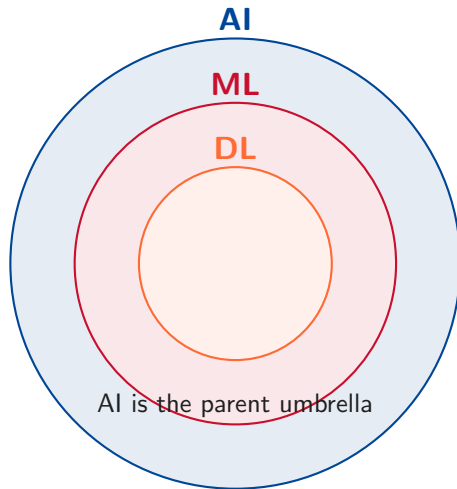
💬 Interaction



AI is a Broad Umbrella

The AI Landscape

- ⚙️ **Machine Learning** – Learn from data
- ⚙️ **Deep Learning** – Neural networks
- 📄 **NLP** – Understand language
- 📷 **Computer Vision** – See and interpret
- ⚙️ **Robotics** – Physical automation

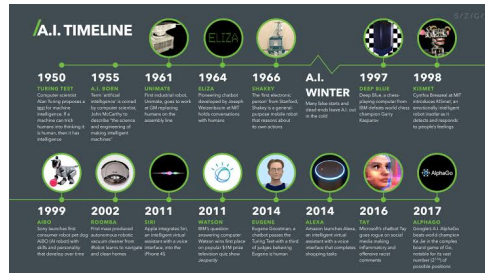


A Brief History of AI

A Brief History of AI

Key Milestones

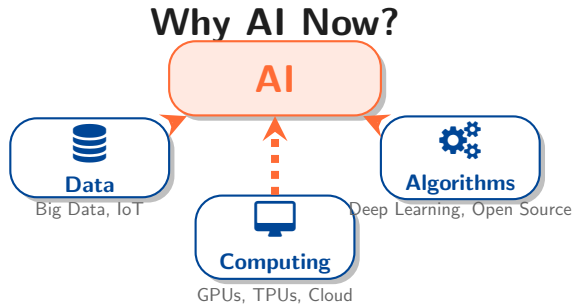
- 1950** Turing Test proposed
- 1956** Dartmouth Conference
- 1960-70s** Expert Systems
- 1980s** Machine Learning emerges
- 1997** Deep Blue beats Kasparov
- 2012** Deep Learning breakthrough
- 2016** AlphaGo beats world champion
- 2020s** Generative AI revolution



Why AI Now?

Three Key Drivers

1. **Data Explosion**
 - Massive digital data
 - Cheap storage
2. **Computing Power**
 - GPUs and TPUs
 - Cloud computing
3. **Algorithm Advances**
 - Deep Learning
 - Open-source frameworks



The convergence of these drivers enabled modern AI

Types of AI

Types of AI – By Capability

Narrow AI (Today)

- Designed for specific tasks
- Most AI today
- **Examples:**
 -  Image recognition
 -  Chatbots
 -  Recommendation systems
 -  OCR systems



Future AI

- **General AI**
 - Human-level intelligence
 - Not yet achieved
- **Super AI**
 - Surpasses human intelligence
 - Theoretical

What is Machine Learning?

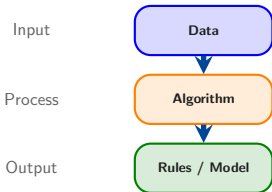
What is Machine Learning?

Machine Learning (ML) is a subset of AI that enables systems to **automatically learn and improve** from experience without explicit programming.

The Core Idea

- Instead of writing rules...
- ...we write algorithms that learn rules
- From **data**, not instructions

How ML Works



The system learns rules from data

Traditional Programming vs Machine Learning

Traditional Programming

Data + Rules = Output

- Programmer writes rules
- Fixed logic
- Explicit instructions



Machine Learning

Data + Output = Rules

- System learns rules
- Adaptive logic
- Implicit learning



Example: Spam Filter

Instead of programming rules for spam, ML learns from examples of spam and not-spam.

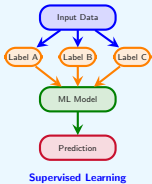
Types of Machine Learning

Types of Machine Learning

Types of Machine Learning

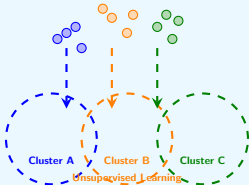
Supervised Learning

- Labeled data
- Predict outcomes
- **Examples:**
 - Classification
 - Regression



Unsupervised Learning

- Unlabeled data
- Find patterns
- **Examples:**
 - Clustering
 - Dimensionality reduction

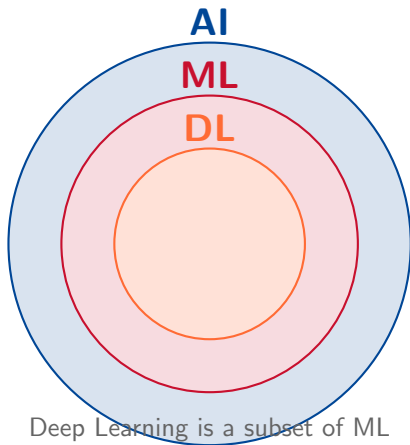


Reinforcement Learning

- Trial and error
- Rewards and punishments
- **Examples:**
 - Game playing
 - Robotics



AI vs ML vs Deep Learning



Deep Learning is a subset of ML
which is a subset of AI

The Relationship

- ⚙️ **AI** – The broadest concept
- </> **ML** – A subset of AI
- ⚙️ **Deep Learning** – A subset of ML

In Simple Terms

- ⚙️ AI = Intelligence
- 🎓 ML = Learning from data
- ⚙️ DL = Learning using brain-inspired networks

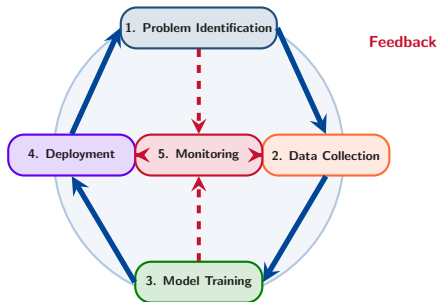
The AI Project Lifecycle

The AI Project Lifecycle

The 5 Stages

1. **Problem Identification**
 - Define the problem
 - Identify success criteria
2. **Data Collection & Preparation**
 - Gather and clean data
 - Handle missing values
3. **Model Selection & Training**
 - Choose algorithms
 - Train and evaluate
4. **Deployment & Integration**
 - Deploy to production
 - Integrate with systems
5. **Monitoring & Maintenance**
 - Track performance
 - Retrain as needed

AI Project Lifecycle



Continuous feedback loop drives improvement

AI in Election Management

AI in Election Management

Why AI in Elections?

- 🕒 **Efficiency** – Automate manual processes
- ✅ **Accuracy** – Reduce human error
- ⚡ **Speed** – Faster processing
- 💡 **Insights** – Data-driven decisions
- 🛡️ **Security** – Detect anomalies
- 👥 **Accessibility** – Better voter services



AI Applications at INEC

Key Application Areas

- **Voter Registration**
 - Biometric verification
 - Duplicate detection
- **Results Management**
 - Anomaly detection
 - Result verification
- **Logistics**
 - Route optimization
 - Resource allocation

More Applications

- **Voter Education**
 - AI chatbots
 - Multilingual support
- **Document Verification**
 - OCR technology
 - Tamper detection
- **Social Media Monitoring**
 - Misinformation detection
 - Sentiment analysis

Specific INEC Use Cases

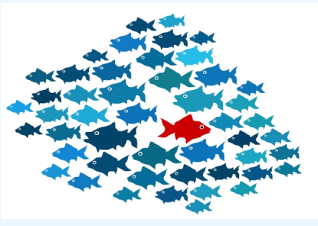
EC8A Tamper Detection

- Computer Vision
- Image comparison
- Pattern recognition



Anomaly Detection

- ML models
- Statistical analysis
- Real-time alerts



Logistics Optimization

- Route optimization
- Predictive planning
- Risk assessment



Responsible AI

Responsible AI – Key Principles

Fairness & Bias

- AI must be unbiased
- Equal treatment for all citizens
- Regular bias audits

Privacy & Security

- Protect voter data
- Secure AI systems
- Data protection compliance

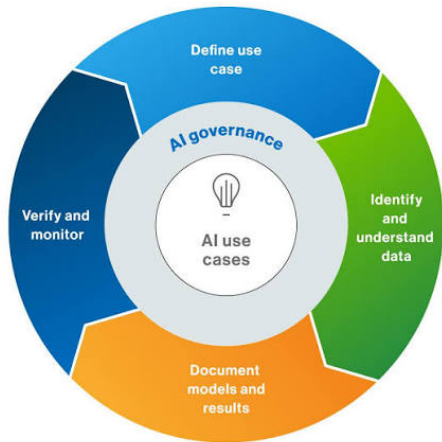
Transparency

- Decisions must be explainable
- Clear documentation
- Explainable AI (XAI)

Accountability

- Clear responsibility
- Human oversight
- Regular audits

AI Governance Framework



Key Questions for INEC

- ✓ How do we ensure AI is fair?
- ✓ Who is accountable for AI decisions?
- ✓ How do we protect voter data?
- ✓ How do we explain AI decisions?
- ✓ How do we monitor AI systems?

Key Takeaways

Key Takeaways

What We Learned

1. **AI** is transforming organizations
2. **ML** is a core component of **AI**
3. **AI** applications are diverse
4. **Ethical AI** is essential
5. **AI** is a tool, not a replacement

💡 Think About

- 👤 How can AI improve **your role** at INEC?
- 🗄️ What **data** would be most valuable for AI?
- ⚖️ How can we ensure AI is **fair** to all?
- 🛡️ What **concerns** do you have about AI?

Resources for Further Learning

Online Courses

-  Coursera: AI for Everyone
-  Kaggle Learn ML
-  DeepLearning.AI

Recommended Reading

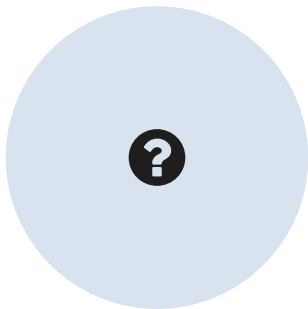
-  *Prediction Machines*
-  *The Master Algorithm*
-  *Human Compatible*

Communities

-  Electoral AI Consortium
-  IEEE AI Ethics Group
-  INEC AI Community of Practice

Thank You

Questions & Discussion



"The future of elections is intelligent."